

Project Proposal: Game Arena — A Benchmark for Comparing AI Model Performance

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Problem Statement & Motivation

Comparing the performance of different AI models (e.g., different reinforcement learning algorithms, or the same algorithm with different hyperparameters/architectures) is often abstract and hard to communicate to non-technical audiences. Benchmarks reported as tables of numbers make it difficult to build intuition for why one model outperforms another. We propose building a **Game Arena**: a platform where multiple AI models play the same game and their performance is measured, compared, and visualized side by side. This makes model comparison tangible and educational; useful both as a teaching tool and as a lightweight benchmark suite for other students or hobbyists testing their own models.

Stakeholder: Students and hobbyists learning reinforcement learning/game AI who want an intuitive, visual way to benchmark their models against known baselines, rather than relying solely on numeric leaderboards.

AI Technique(s) & Rationale

- **Primary technique:** Reinforcement learning (e.g., Q-learning, and potentially a deep RL method like DQN if time allows) applied to a simple game environment (e.g., a custom Mario-style level, Snake, or an existing open-source game with a suitable API).
- **Why this technique:** The course explicitly lists RL applied to simple games (Snake, Connect-4, Tic-Tac-Toe) as a suggested direction, and it naturally supports quantitative evaluation (win rate, score, episodes-to-convergence) which the rubric requires.
- **Comparison dimension:** We will train/evaluate at least 2–3 distinct model types or configurations (e.g., Q-learning vs. a random baseline vs. a deep RL agent, or the same architecture under different hyperparameters) to generate meaningful comparative results rather than a single model in isolation.
- **Stretch goal:** A lightweight web front-end allowing external users to upload or plug in their own model/agent and see it benchmarked against our existing leaderboard — this would extend the technique set to include a basic web/API layer alongside the core RL work.

Data & Tools Needed

- Game environment: either a custom-built simple game (e.g., Python + Pygame) or an existing open-source environment (e.g., OpenAI Gym / Gymnasium environments such as CartPole, or a simple existing Snake/Connect-4 implementation)
- Python, with RL libraries such as gymnasium, stable-baselines3, or a from-scratch Q-learning implementation depending on scope.
- Visualization tools: matplotlib/plotly for performance dashboards (win rate curves, score over episodes, comparison bar charts).
- (Stretch) Basic web framework (e.g., Flask or Streamlit) if we pursue the public-facing benchmark site.

Team Roles

- **Gerard:** Game environment setup / integration with chosen open-source game or custom build
- **Martha:** RL model implementation and training (baseline + comparison models)
- **Kofi:** Evaluation framework and visualization/dashboard
- **Lilly:** Report writing, ethics audit, and (if pursued) the web front-end for the stretch goal

Milestone Plan (Weeks 10–13)

Week	Goal
10	Finalize game choice and confirm technical feasibility; set up GitHub repo with README skeleton; scope which model types will be compared
11	Development Sprint 1 — implement baseline game environment + first working model (e.g., random agent + simple Q-learning agent)
12	Development Sprint 2 — train/tune comparison models; build initial evaluation metrics and visualizations; informal demo at lab
13	Finalize evaluation results; conduct ethics & fairness audit (e.g., fairness of comparisons across model types, transparency of benchmark methodology); polish README; build slide deck; (stretch) basic web interface if time allows